
Day 5

Tasks

1. List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each.

Main Components Inside a Desktop PC

No.	Component	Short Description
1.	Motherboard	The main circuit board of the PC that connects and allows communication between the CPU, RAM, storage, GPU, and other components.
2.	CPU (Processor)	The main processing unit that executes instructions and performs calculations required to run programs and the operating system.
3.	CPU Cooler	Removes heat produced by the CPU using a heatsink and fan or a liquid-cooling system.
4.	RAM (Memory)	Provides temporary, high-speed storage for data and programs that are currently being used by the computer.
5.	GPU (Graphics Card)	Processes graphics and video output. It is especially important for gaming, video editing, and other graphics-intensive tasks.
6.	Storage Drive (HDD/SSD/NVMe)	Stores the operating system, applications, games, documents, and other files permanently.
7.	Power Supply Unit (PSU)	Converts electricity from the wall outlet into the appropriate voltages required by the PC components.
8.	Case/Cabinet	Houses and protects the internal components while providing proper airflow and space for installation.
9.	Case Fans	Move air through the cabinet to remove hot air and bring in cooler air, helping prevent overheating.
10.	Optical Drive (if present)	Reads or writes CDs and DVDs. It is less common in modern desktop PCs.
11.	Expansion Cards (if present)	Additional cards installed in PCIe slots to provide features such as Wi-Fi, sound, network connectivity, or specialized functions.
12.	CMOS Battery	A small battery on the motherboard that helps maintain BIOS/UEFI settings and the system clock when the PC is powered off.
13.	SATA/Data & Power Cables	Connect storage drives and other components to the motherboard and power supply, allowing data transfer and power delivery.

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2. Take clear photos or draw labelled diagrams of each step as you disassemble a desktop PC, and organize them in a document with captions explaining what you did at each step.

Hint: Include steps like unplugging power, removing side panel, disconnecting cables, and removing RAM, HDD/SSD, and motherboard.

Desktop PC Disassembly – Step-by-Step

Step 1: Shut Down and Unplug the PC:

The computer was completely shut down, and the power cable was disconnected from the power supply and wall socket. All other external cables and peripherals were also removed.



Step 2: Remove the Side Panel:

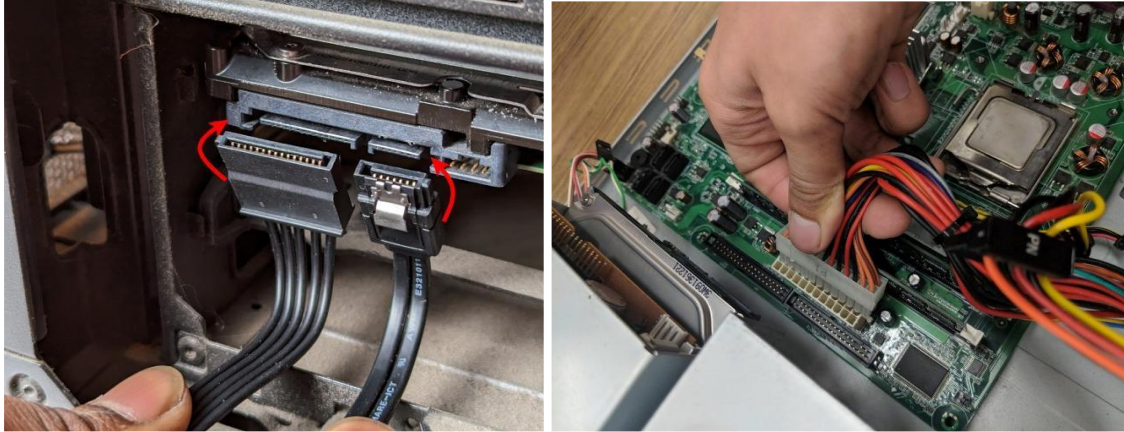
The screws holding the side panel were removed, and the panel was carefully taken off to access the internal components of the desktop PC.



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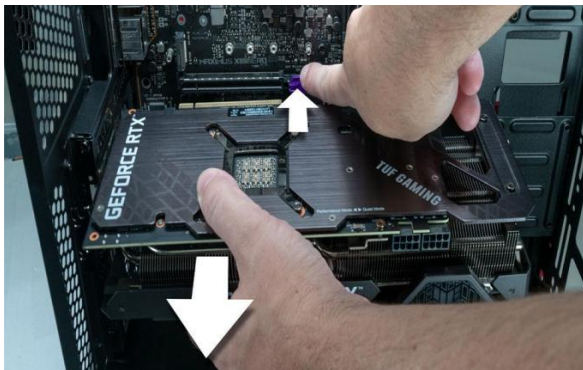
Step 3: Disconnect Internal Cables:

The power and data cables connected to the motherboard, storage drives, GPU, and other components were carefully disconnected.



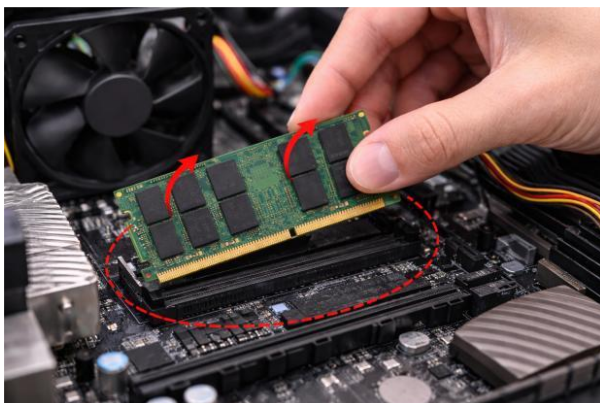
Step 4: Remove the Graphics Card (GPU):

The GPU was removed by disconnecting its power cable, removing the bracket screw, and carefully releasing it from the PCIe slot on the motherboard.



Step 5: Remove the RAM:

The locking clips on both sides of the RAM slot were released, and the RAM modules were carefully removed from the motherboard.



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Step 6: Remove the HDD/SSD:

The storage drive was disconnected from its SATA/data and power cables. The screws or mounting mechanism were then released, and the HDD/SSD was removed from its bay.



Step 7: Remove the Power Supply Unit (PSU):

After ensuring that all power cables were disconnected, the screws securing the PSU to the cabinet were removed, and the power supply was taken out.



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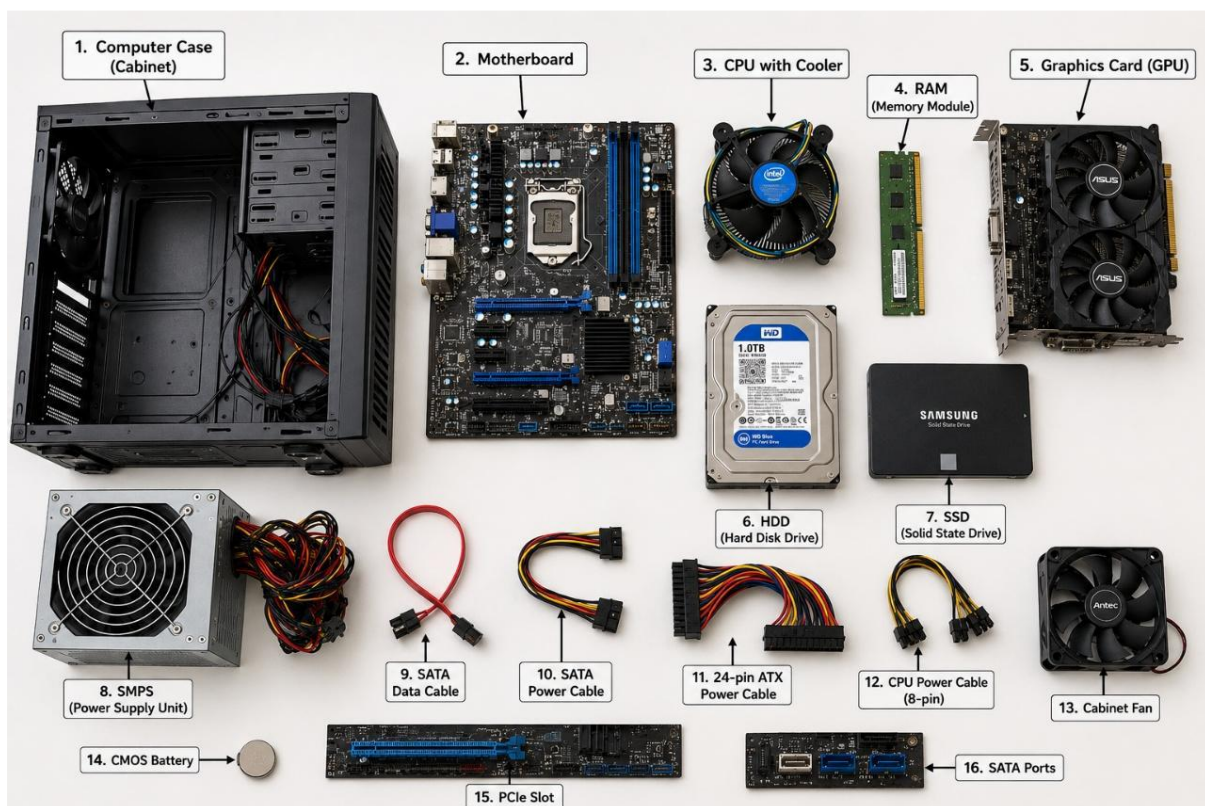
Step 8: Remove the Motherboard:

The remaining motherboard connections were checked and disconnected. The motherboard screws were removed, and the motherboard was carefully lifted out of the cabinet.



Step 9: Final Disassembled Components:

After completing the disassembly, the major components were placed separately and identified, including the motherboard, CPU, RAM, GPU, storage drive, CPU cooler, PSU, and cables.



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Safety Note

Always disconnect the PC from electrical power before opening the cabinet. Handle components by their edges, avoid touching electrical contacts, and keep screws organized so they can be used during reassembly.

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3.Reassemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters.

Desktop PC Reassembly – Step-by-Step

Step 1: Install the Motherboard

The motherboard was carefully placed inside the cabinet and aligned with the motherboard standoffs and I/O opening. The screws were tightened to secure it properly because all other components connect to the motherboard.

Step 2: Install the CPU

The CPU was placed into the correct socket on the motherboard and secured using the retention mechanism. Correct installation is important because the CPU must be properly aligned before installing the cooler.

Step 3: Install the CPU Cooler

The CPU cooler was installed on top of the processor and connected to the CPU_FAN header on the motherboard. The cooler is necessary to remove heat generated by the CPU during operation.

Step 4: Install the RAM

The RAM modules were inserted into the appropriate memory slots until the locking clips clicked into place. RAM needs to be installed securely so the computer can access temporary data while running.

Step 5: Install the GPU

The graphics card was inserted into the appropriate PCIe slot and secured to the cabinet with a screw. If required, its power cable was connected to the power supply.

Step 6: Install the HDD/SSD

The storage drive was placed in its appropriate mounting area and secured. The SATA/data cable was connected to the motherboard and the power cable was connected to the PSU.

Step 7: Install the Power Supply

The PSU was positioned inside the cabinet and secured with screws. The main motherboard power cable, CPU power cable, GPU power cable, and storage power cables were connected as required.

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Step 8: Connect the Front-Panel Cables

The cabinet's power button, reset button, LEDs, and front USB/audio cables were connected to their corresponding motherboard headers. These connections allow the cabinet's front controls and ports to function.

Step 9: Connect Case Fans

The case fans were connected to the appropriate motherboard fan headers or power connectors. Proper fan installation helps maintain airflow and prevents overheating.

Step 10: Check All Connections

All components and cables were checked to make sure they were properly installed and securely connected. Loose connections can prevent the PC from starting or cause hardware problems.

Step 11: Close the Cabinet

After confirming that everything was correctly installed, the side panel was placed back and secured with screws.

Step 12: Test the PC

The power cable and external peripherals were reconnected, and the computer was switched on. The system was checked to confirm that it successfully booted and recognized the installed hardware.

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4. Create a checklist of safety precautions and tools needed before starting the PC assembly/disassembly process, and explain why each is important.

Tools Needed

☒	Tool	Why it is needed
☐	Screwdriver	To remove and tighten screws.
☐	Anti-static wrist strap	Protects PC parts from static electricity.
☐	Small box/container	Keeps screws safe and organized.
☐	Soft cloth	Used to clean dust from components.
☐	Compressed air	Helps remove dust from fans and other parts.

Safety Precautions

☒	Safety Check	Why it is important
☐	Shut down the PC	Prevents damage while working on it.
☐	Unplug the power cable	Protects against electric shock and short circuits.
☐	Turn off the PSU switch	Provides extra electrical safety.
☐	Work on a clean and dry table	Prevents dust, water, and other damage.
☐	Avoid static electricity	Protects sensitive parts like RAM and the motherboard.
☐	Handle parts carefully	Prevents damage to components and connectors.
☐	Keep screws safely	Prevents screws from getting lost.
☐	Do not force components	Prevents damage when installing RAM, GPU, or other parts.
☐	Check all connections before turning on	Makes sure everything is connected correctly and safely.